

- Mild cognitive impairment
- Chronic pain

How the Drug Works

- * Is a low to moderate affinity noncompetitive (open-channel) NMDA receptor antagonist, which binds preferentially to the NMDA receptor-operated cation channels
- Presumably interferes with the postulated persistent activation of NMDA receptors by excessive glutamate release in Alzheimer disease

How Long Until It Works

- Memory improvement is not expected and it may take months before any stabilization in degenerative course is evident

If It Works

- May slow progression of disease, but does not reverse the degenerative process

If It Doesn't Work

- Consider adjusting dose, switching to a cholinesterase inhibitor or adding a cholinesterase inhibitor
- Reconsider diagnosis and rule out other conditions such as depression or a dementia other than Alzheimer disease

Best Augmenting Combos for Partial Response or Treatment Resistance

- * Atypical antipsychotics to reduce behavioral disturbances
- * Antidepressants if concomitant depression, apathy, or lack of interest
- * May be combined with cholinesterase inhibitors
- Divalproex, carbamazepine, or oxcarbazepine for behavioral disturbances

Tests

- None for healthy individuals

Side Effects

How Drug Causes Side Effects

- Presumably due to excessive actions at NMDA receptors

Notable Side Effects

- Dizziness, headache
- Constipation



Life-Threatening or Dangerous Side Effects

- Seizures (rare)

Weight Gain



- Reported but not expected

Sedation



- Reported but not expected
- Fatigue may occur

What to Do About Side Effects

- Wait
- Wait
- Wait
- Consider lowering dose or switching to a different agent

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 10 mg twice daily
- 28 mg once daily (extended-release)

- Namzaric extended-release (combination memantine/donepezil) 7 mg/10 mg, 14 mg/10 mg, 21 mg/10 mg, 28 mg/10 mg

Dosage Forms

- Tablet 5 mg, 10 mg
- Oral solution 2 mg/mL
- Extended-release capsule 7 mg, 14 mg, 21 mg, 28 mg

How to Dose

- Initial 5 mg/day; can increase by 5 mg each week; doses over 5 mg should be divided; maximum dose 10 mg twice daily
- Extended-release: initial 7 mg once daily; can increase by 7 mg each week; maximum dose 28 mg once daily
- Namzaric, for patients already on donepezil 10 mg once daily: initial dose 7 mg/10 mg once daily in the evening; increase weekly in 7 mg increments; maximum dose 28 mg/10 mg
- Namzaric, for patients already on memantine (10 mg twice per day or 28 mg extended-release once per day) and donepezil 10 mg once daily: can switch to 28 mg/10 mg once daily in the evening



Dosing Tips

* Despite very long half-life, is generally dosed twice daily, although some data suggest once daily is safe and tolerable

- Both the patient and the patient's caregiver should be instructed on how to dose memantine since patients themselves have moderate to severe dementia and may require assistance
- * Memantine is unlikely to affect pharmacokinetics of acetylcholinesterase inhibitors
- Absorption not affected by food
- Namzaric can be taken with or without food, whole or sprinkled on applesauce; do not divide, chew, or crush

Overdose

- No fatalities have been reported; restlessness, psychosis, visual hallucinations, sedation, stupor, loss of consciousness

Long-Term Use

- Drug may lose effectiveness in slowing degenerative course of Alzheimer disease after 6 months

Habit Forming

- No

How to Stop

- No known withdrawal symptoms
- Theoretically, discontinuation could lead to notable deterioration in memory and behavior which may not be restored when drug is

restarted or a cholinesterase inhibitor is initiated

Pharmacokinetics

- Little metabolism; mostly excreted unchanged in the urine
- Terminal elimination half-life approximately 60–80 hours
- Minimal inhibition of CYP450 enzymes



Drug Interactions

- No interactions with drugs metabolized by CYP450 enzymes
- Drugs that raise the urine pH (e.g., carbonic anhydrase inhibitors, sodium bicarbonate) may reduce elimination of memantine and raise plasma levels of memantine
- * No interactions with cholinesterase inhibitors



Other Warnings/Precautions

- * Use cautiously if coadministering with other NMDA antagonists such as amantadine, ketamine, and dextromethorphan

Do Not Use

- If there is a proven allergy to memantine

Special Populations

Renal Impairment

- No dose adjustment in mild or moderate impairment
- Reduce dose in severe impairment
- Severe renal impairment: the recommended maintenance dose for Namzaric is 14 mg/10 mg once daily in the evening

Hepatic Impairment

- Not likely to require dosage adjustment

Cardiac Impairment

- Not likely to require dosage adjustment

Elderly

- Pharmacokinetics similar to younger adults



Children and Adolescents

- Memantine use has not been studied in children or adolescents



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001

- Controlled studies have not been conducted in pregnant women
- Animal studies do not show adverse effects
- * Not recommended for use in pregnant women or women of childbearing potential

Breast Feeding

- Unknown if memantine is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed
- Memantine is not recommended for use in nursing women

The Art of Psychopharmacology

Potential Advantages

- In patients with more advanced Alzheimer disease

Potential Disadvantages

- Unproven to be effective in mild to moderate Alzheimer disease
- Patients who have difficulty taking a medication twice daily

Primary Target Symptoms

- Memory loss in Alzheimer disease
- Behavioral symptoms in Alzheimer disease

- Memory loss in other dementias

Pearls

* Memantine's actions are somewhat like the natural inhibition of NMDA receptors by magnesium, and thus memantine is a sort of “artificial magnesium”

- Theoretically, NMDA antagonism of memantine is strong enough to block chronic low-level overexcitation of glutamate receptors associated with Alzheimer disease, but not strong enough to interfere with periodic high level utilization of glutamate for plasticity, learning, and memory
- Structurally related to the antiparkinsonian and anti-influenza agent amantadine, which is also a weak NMDA antagonist
- * Memantine is well tolerated with a low incidence of adverse effects
- Antagonist actions at 5HT₃ receptors have unknown clinical consequences but may contribute to low incidence of gastrointestinal side effects
- A fixed-dose combination of memantine extended-release and donepezil has been approved for the treatment of moderate to severe Alzheimer dementia in patients stabilized on memantine and donepezil
- Treat the patient but ask the caregiver about efficacy
- Delay in progression of Alzheimer disease is not evidence of disease-modifying actions of NMDA antagonism

- May or may not be effective in vascular dementia
- Under investigation for dementia associated with HIV/AIDS
- May or may not be effective in chronic neuropathic pain
- * Theoretically, could be useful for any condition characterized by moderate overactivation of NMDA glutamate receptors (possibly neurodegenerative conditions or even bipolar disorder, anxiety disorders, or chronic neuropathic pain), but this is not proven

Suggested Reading

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Mobius HJ . Memantine: update on the current evidence . *Int J Geriatr Psychiatry* 2003 ;**18** (Suppl 1):S47 –54.

Sani G , Serra G , Kotzalidis GD , et al. The role of memantine in the treatment of psychiatric disorders other than the dementias: a review of current preclinical and clinical evidence . *CNS Drugs* 2012 ;**26** (8):663–90 .

Tariot PN , Federoff HJ . Current treatment for Alzheimer disease and future prospects . *Alzheimer Dis Assoc Disord* 2003 ;**17** (Suppl 4):S105 –13.

Methylfolate (L)

Therapeutics

Brands

- Deplin
- *see index for additional brand names*

Generic?

No

Class

- Medical food (bioavailable form of folate)
- Trimonoamine modulator

Commonly Prescribed for

(bold for FDA approved as medical food indications)

- **Suboptimal folate levels in depressed patients (adjunct to antidepressant)**
- **Hyperhomocysteinemia in schizophrenia patients (adjunct to antipsychotic)**
- Enhancement of antidepressant response at the initiation of treatment

- Cognitive or mood symptoms in patients with MTHFR (methylene tetrahydrofolate) deficiency

How the Drug Works

- Folate is a water-soluble B vitamin (B9) that is essential for cell growth/reproduction, breakdown/utilization of proteins, formation of nucleic acids, and other functions
- L-methylfolate, or 6-(S)-5-methyl-tetrahydrofolate, is derived from folate and is the form that enters the brain and works directly as a methyl donor and monoamine synthesis modulator
- That is, it regulates tetrahydrobiopterin (BH4), a critical enzyme cofactor for trimonoamine neurotransmitter synthesis
- Methyl donor for DNA methylation and thus an epigenetic regulator

How Long Until It Works

- Onset of therapeutic actions in depression is usually not immediate, but often delayed 2–4 weeks
- If it is not working within 6–8 weeks for depression, it may require a dosage increase or it may not work at all
- May continue to work for many years to prevent relapse of symptoms

If It Works

- The goal of treatment for depression is complete remission of current symptoms as well as prevention of future relapses
- Treatment most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine is stopped
- Continue treatment until all symptoms are gone (remission)
- Once symptoms are gone, continue treating for 1 year for the first episode of depression
- For second and subsequent episodes of depression, treatment may need to be indefinite

If It Doesn't Work

- Many patients with depression have only a partial response where some symptoms are improved but others persist (especially insomnia, fatigue, and problems concentrating)
- Other patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Consider increasing dose, switching to another agent, or adding an appropriate augmenting agent
- Consider psychotherapy
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)
- Some patients may experience apparent lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require

antidepressant discontinuation and a switch to a mood stabilizer



Best Augmenting Combos for Partial Response or Treatment Resistance

- L-methylfolate is itself an adjunct to standard treatments for depression or schizophrenia at the initiation of treatment or to augment a partial response

Tests

- Baseline folate levels (serum levels for recent folate intake; red blood cell or CSF levels for long-term folate levels)
- Baseline homocysteine levels (reciprocal relationship with folate levels; high homocysteine levels may be more sensitive in detected folate deficiency than folate levels themselves in some patients)
- May monitor folate levels for patients taking an agent capable of affecting folate metabolism, absorption, or degradation
- May consider genotyping for deficient folate synthesis via MTHFR T alleles or MTHFD1 A alleles

Side Effects

How Drug Causes Side Effects

- L-methylfolate does not typically cause side effects

Notable Side Effects

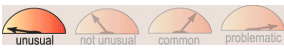
- L-methylfolate does not typically cause side effects



Life-Threatening or Dangerous Side Effects

- Theoretically, rare induction of mania or suicidal ideation and behavior (suicidality)

Weight Gain



- Reported but not expected

Sedation



- Reported but not expected

What to Do About Side Effects

- Wait
- Lower the dose or administer in divided doses
- Switch to another drug

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent
- Activation and agitation may represent the induction of a bipolar state, especially a mixed dysphoric bipolar II condition sometimes

associated with suicidal ideation, and require the addition of lithium, a mood stabilizer or an atypical antipsychotic, and/or discontinuation of l-methylfolate or the primary antidepressant

Dosing and Use

Usual Dosage Range

- 7.5–15 mg/day

Dosage Forms

- Tablet 7.5 mg, 15 mg

How to Dose

- Initial 7.5–15 mg/day
- Doses above 15 mg/day should be administered in divided doses



Dosing Tips

- Can be taken with or without food
- L-methyltetrahydrofolate was shown to be 7 times more bioavailable than folic acid
- That means 7.5 mg of the active l enantiomer of methylfolate may be equivalent to 52 mg of folate (usual dose of folate is 100 ug to 1.0 mg)

- If intolerable anxiety, insomnia, agitation, akathisia, or activation occur upon either dosing initiation or discontinuation, consider the possibility of activated bipolar disorder and switch to a mood stabilizer or an atypical antipsychotic

Overdose

- Doses up to 90 mg/day of methylfolate (45 mg l-methylfolate) have been studied without any additional adverse events
- L-methylfolate is generally regarded as safe
- A toxic dose of l-methylfolate is not known at this time

Long-Term Use

- Safe

Habit Forming

- No

How to Stop

- Taper not necessary

Pharmacokinetics

- Mean elimination half-life approximately 3 hours for d,l-methylfolate

- L-methylfolate is naturally stored in most cells and used by the body when needed; therefore, l-methylfolate may not follow typical drug pharmacokinetic patterns

Drug Interactions

- L-methylfolate may reduce plasma levels of certain anticonvulsants, including phenytoin, carbamazepine, fosphenytoin, phenobarbital, primidone, or valproate
- L-methylfolate may reduce plasma levels of pyrimethamine
- Patients taking folate-lowering drugs (e.g., anticonvulsants, cholestyramine, colestipol, cycloserine, aminopterin, methotrexate, sulfasalazine, pyrimethamine, triamterene, trimethoprim, isotretinoin, fluoxetine, nonsteroidal anti-inflammatory drugs (NSAIDs), methylprednisolone, pentamidine, or who smoke or drink heavily may require higher doses of l-methylfolate

Other Warnings/Precautions

- Folic acid may mask symptoms of B12 deficiency (e.g., pernicious anemia), although this may be less likely with l-methylfolate
- Use with caution in patients with bipolar disorder unless treated with concomitant mood-stabilizing agent
- Monitor patients for activation of suicidal ideation, especially children and adolescents

- Folic acid, when administered in doses above 800 $\hat{1}\frac{1}{4}$ g, may increase the amount of unmetabolized folic acid, which has been linked to accelerated growth of existing neoplasms in the colon; l-methylfolate may be less likely than folic acid to accelerate the growth of existing neoplasms

Do Not Use

- If there is a proven allergy to folate or folic acid

Special Populations

Renal Impairment

- Dose adjustment not necessary

Hepatic Impairment

- Dose adjustment not necessary

Cardiac Impairment

- Dose adjustment not necessary

Elderly

- Dose adjustment not necessary



Children and Adolescents

- Use with caution, observing for activation of known or unknown bipolar disorder and/or suicidal ideation, and inform parents or guardians of this risk so they can help observe child or adolescent patients
- Safety and efficacy have not been established

Pregnancy

- No controlled studies in humans or animals
- Controlled studies of folic acid at recommended doses have failed to demonstrate risk to the fetus
- There are no studies of folic acid at high doses
- Because pregnant women are advised to take folic acid or prenatal vitamins that contain folic acid, it is important to ask the patient about any supplements or vitamins she may be taking and consider this when deciding whether to prescribe l-methylfolate

Breast Feeding

- Some drug is found in mother's breast milk

The Art of Psychopharmacology

Potential Advantages

- Patients who need efficacy greater than an antidepressant alone at the initiation of treatment

- Patients with partial or inadequate response to antidepressants
- Patients who cannot tolerate other antidepressants

Potential Disadvantages

- Patients with adequate folate levels

Primary Target Symptoms

- Depressed mood
- Cognitive symptoms

Pearls

- Numerous studies suggest that low plasma, red blood cell, and/or CSF levels of folate may be associated with depression in some patients
- Treatment with l-methylfolate seems to be safe, has few if any side effects, and is generally less expensive than augmenting with a second branded antidepressant or atypical antipsychotic
- L-methylfolate is able to cross the blood–brain barrier and support the synthesis of monoamines
- Early studies suggest that those with obesity may be better responders
- Early studies suggest that those with genetic polymorphisms reducing the formation of l-methylfolate may be better responders

Suggested Reading

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Fava M , Mischoulon D . Folate in depression: efficacy, safety, differences in formulations, and clinical issues . *J Clin Psychiatry* 2009 ;**70** (Suppl 5):S12 –17.

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Miller AL . The methylation, neurotransmitter, and antioxidant connections between folate and depression . *Altern Med Rev* 2008 ;**13** (3):216–26 .

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Stahl SM . L-methylfolate: a vitamin for your monoamines . *J Clin Psychiatry* 2008 ;**69** (9):1352 –3.

Methylphenidate (D)

Therapeutics

Brands

- Focalin
- Focalin XR
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: dopamine, norepinephrine reuptake inhibitor and releaser (DN-RIRe)
- Stimulant

Commonly Prescribed for

(bold for FDA approved)

- **Attention deficit hyperactivity disorder (ADHD) in children ages 6–17 (Focalin, Focalin XR) and in adults (Focalin XR)**
- Narcolepsy
- Treatment-resistant depression

How the Drug Works

- * Increases norepinephrine and especially dopamine actions by blocking their reuptake
- Enhancement of dopamine and norepinephrine actions in certain brain regions (e.g., dorsolateral prefrontal cortex) may improve attention, concentration, executive function, and wakefulness
- Enhancement of dopamine actions in other brain regions (e.g., basal ganglia) may improve hyperactivity
- Enhancement of dopamine and norepinephrine in yet other brain regions (e.g., medial prefrontal cortex, hypothalamus) may improve depression, fatigue, and sleepiness

How Long Until It Works

- Onset of action can occur 30 minutes post-administration
- Can take several weeks to attain maximum therapeutic benefit

If It Works (for ADHD)

- The goal of treatment of ADHD is reduction of symptoms of inattentiveness, motor hyperactivity, and/or impulsiveness that disrupt social, school, and/or occupational functioning
- Continue treatment until all symptoms are under control or improvement is stable and then continue treatment indefinitely as long as improvement persists
- Reevaluate the need for treatment periodically

- Treatment for ADHD begun in childhood may need to be continued into adolescence and adulthood if continued benefit is documented

If It Doesn't Work (for ADHD)

- Consider adjusting dose or switching to a formulation of d,l-methylphenidate or to another agent
 - Consider behavioral therapy
 - Consider the presence of noncompliance and counsel patient and parents
 - Consider evaluation for another diagnosis or for a comorbid condition (e.g., bipolar disorder, substance abuse, medical illness, etc.)
- * Some ADHD patients and some depressed patients may experience lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require either augmenting with a mood stabilizer or switching to a mood stabilizer



Best Augmenting Combos for Partial Response or Treatment Resistance

- * Best to attempt other monotherapies prior to augmenting
- For the expert, can combine immediate-release formulation of d-methylphenidate with a sustained-release formulation of d-methylphenidate for ADHD